

And here's the most important concept for YOU

You **do not need to become a civil engineer** to build this project.

We need to understand the documents at **three levels**:

Level 1 — Domain

What is this document about?

S-1 → materials

S-2 → railway sleepers

S-3 → concrete construction

Level 2 — Information

What does the document contain?

Materials, standards, processes, tests, requirements, conditions, approvals, acceptance criteria.

Level 3 — Relationships

How are those things connected?

```
Material
  ↓
must conform to
  ↓
Standard
  ↓
tested by
  ↓
Test
  ↓
approved by
  ↓
Engineer
```

Level 3 is where our GraphRAG project lives.

One more VERY important observation

The three PDFs also demonstrate why our system **cannot be a basic PDF chatbot**.

We have:

S-1: scanned pages + tables → requires OCR/table understanding.

S-2: amendments/corrigenda → requires temporal/version relationships.

S-3: conditions + requirements + acceptance criteria → requires relationship/multi-hop reasoning.

That gives us **three real technical challenges** instead of artificially adding features.

So our mental model should now be:

PDF is not just text.

It's a collection of:

Entities + Requirements + Conditions + Relationships + Evidence + Versions

And our job is to convert:

```
PDF
↓
Information
↓
Relationships
↓
Knowledge Graph
↓
Graph + Vector Retrieval
↓
Reasoning
↓
Evidence-grounded Answer
```